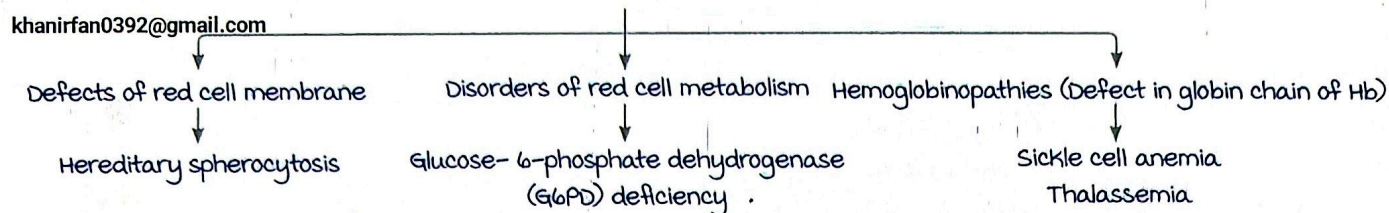


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CONGENITAL HEMOLYTIC ANEMIA

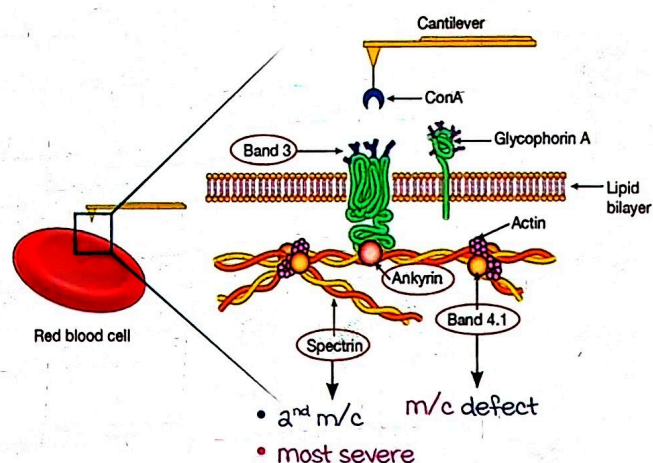
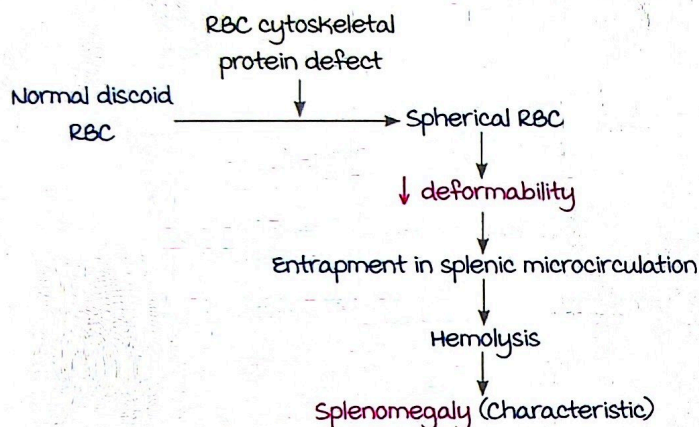


Hereditary Spherocytosis

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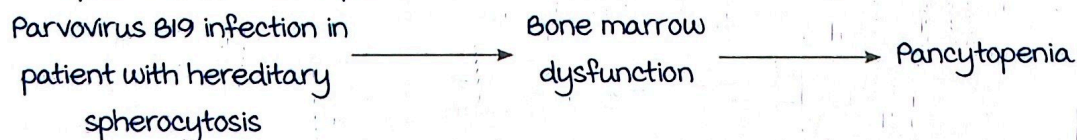
Autosomal dominant inheritance.

PATHOGENESIS :



CLINICAL FEATURES :

- **Variable onset** of anaemia & jaundice (D/t ↑ lysis of RBCs).
 - 50% : Neonatal jaundice.
- Splenomegaly.
- Family history +.
- Pigment gallstones (↑ bilirubin production).
- Aplastic crisis : Acute presentation.



INVESTIGATIONS :**1. Peripheral smear :**

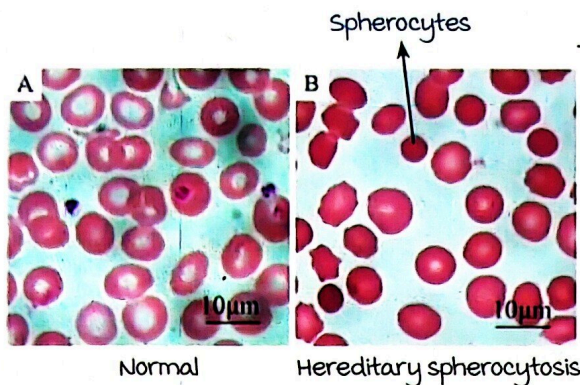
- Lack of central pallor in spherocytes.

Small surface area of RBC is lost
during shape change

↓
Cellular dehydration

↑ Concentration of Hb inside the RBC

↓
Hyperchromia

**2. Red cell indices :**

- ↑ **MCHC** (Characteristic).

3. Screening tests :

- Osmotic fragility : **Increased**

↑ Permeability to Na^+ in spherocytes

↓
↑ $\text{Na}^+/\text{H}_2\text{O}$ transport into RBC

↓
↑ swelling & lysis at lower concentration of hypotonic saline

- Incubated osmotic fragility :

- Patients blood is incubated at 37°C for 24 hours.
- Improves sensitivity of testing.

Osmotic fragility is seen in other hemolytic disorders.

4. Confirmatory tests :

Flow cytometry : ↓ bending of Eosin-5-maleimide (EMA) to proteins.

molecular testing : Deletion of mutations in RBCs.

MANAGEMENT :**1. Lifelong folic acid supplementation :**

- D/t megaloblastic erythropoiesis from chronic hemolysis.
- Given to all children with chronic hemolytic anemia's.

2. Splenectomy :

- **Best** treatment modality.
- ↓ lysis → ↑ lifespan of RBC's → ↓ severity of anemia.
- Laparoscopic **partial splenectomy** preferred.
- Concomitant cholecystectomy if gall stones present.

Consequences of splenectomy :

Absence of splenic macrophages

Risk of infection by encapsulated bacteria (Especially if age < 6 yrs) :

- Streptococcus pneumoniae.
- Haemophilus influenzae type b.
- Meningococcus.

Precautions :

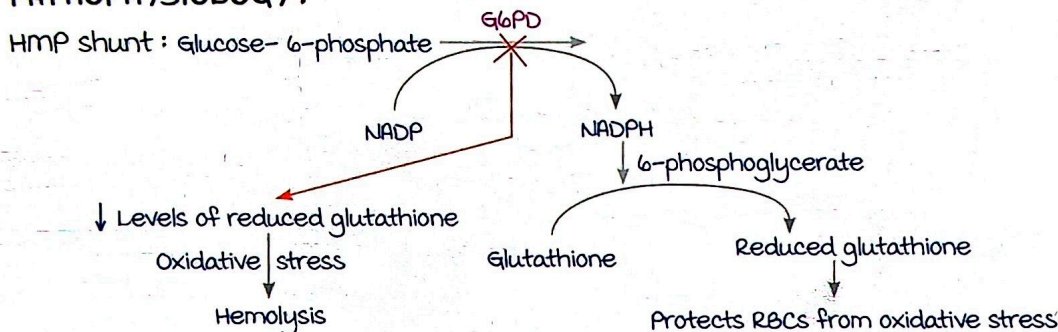
- Surgery > 6 years of age.
- Immunization.
- Penicillin for 2 years post sx.

G6PD Deficiency

00:18:13

- m/c RBC enzyme deficiency.
- X-linked recessive.
- Commonly seen in Africans, Mediterranean's & Asians.

PATHOPHYSIOLOGY :



CLINICAL FEATURES :

- Variable onset
 - Early : Neonatal jaundice.
 - Late : Episodic hemolysis.

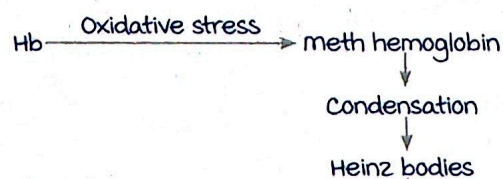
Risk factors for episodic hemolysis :

- Fever.
- Fava beans.
- Drugs (Starts 2-3 days after exposure, self-limiting) :
 - Primaquine.
 - Nalidixic acid.
 - Nitrofurantoin.
 - Phenazopyridine.
 - Sulfonamides.

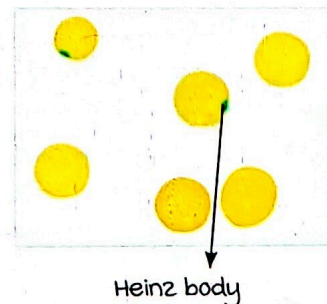
INVESTIGATIONS :

I. Peripheral smear :

- Heinz bodies :



- Bite cells
 - Engulfed by splenic macrophages



Heinz body

a. Enzyme assay in RBCs : **Confirmatory test.**

- ↓ enzyme levels in RBCs.

MANAGEMENT :

Supportive :

- most episodes are self-limiting.
- Avoidance of implicated drugs.

----- Active space -----

Haemoglobinopathies

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SICKLE CELL ANEMIA :

Autosomal recessive.

High incidence in **tribal populations** of madhya Pradesh & maharashtra.

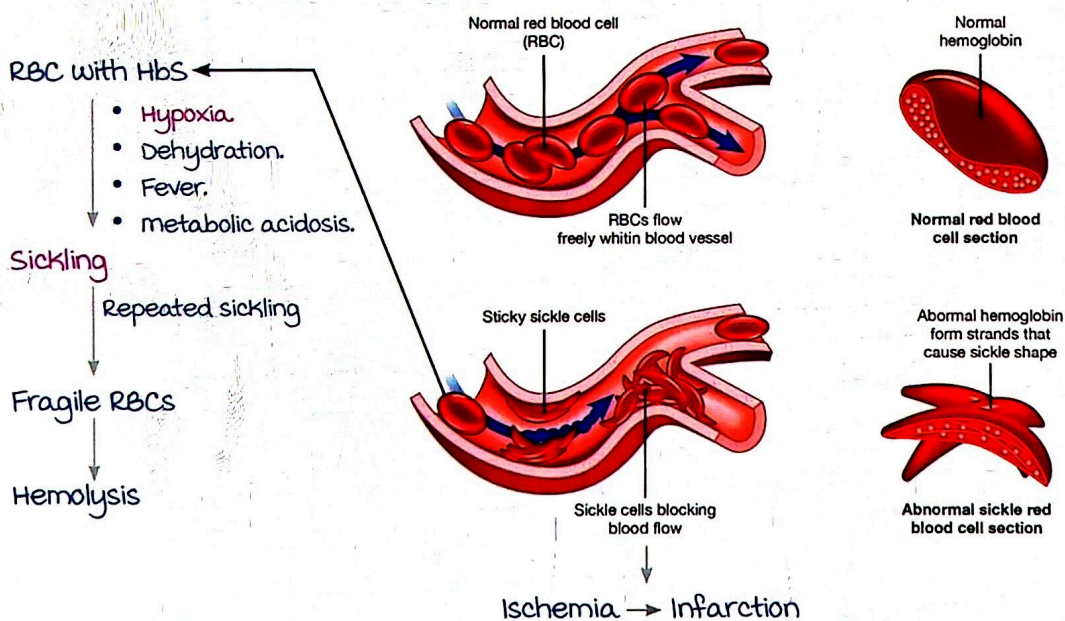
Pathogenesis :

mutation in gene for **β -globin** chain of hemoglobin

↓
Glutamate replaced by valine at 6th position

Production of defective Hb : HbS

(Normal Hb : HbA)



Clinical features :

manifests > 6 months (HbF predominates for 6 months → Protective).

- Icterus.
- Pallor.
- Vaso-occlusive crisis : **Painful episodes** (Characteristic).
- Dactylitis/hand and foot syndrome : Swelling of finger and toes. manifests in infancy.

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Haematology

----- Active space -----

- Acute chest syndrome : Fever & breathing difficulty. Resembles pneumonia. Chest X-ray shows new-onset infiltrates.
- Stroke episodes.

Sickling of cells in spleen :

- Repeated infarcts → Autosplenectomy by 6 years of age (Functional hyposplenism) → Recurrent infection by encapsulated bacteria.
- Venous blockade → Acute pooling of blood → ≥2% Hb reduction → Pallor, shock.

Aplastic crisis : Pancytopenia (Precipitated by parvovirus B19 infection).

Investigations :

1. Peripheral smear : Sickled RBCs.
2. Screening test :
 - **Sickling test** : Induction of sickling by adding 2% sodium metabisulphite.
3. Hb electrophoresis :
 - Initial, best diagnostic test.
4. High-Performance Liquid Chromatography (HPLC) :
 - **Gold standard test.**
 - Detects & quantifies HbS.

Note :

Sickle cell trait :

- HbA + HbS → Less severe manifestations.

MANAGEMENT :

- Good hydration.
- Analgesics for pain episodes.
- PRBC transfusions : Aplastic crisis & sequestration crisis.
 - To ↑ O_2 carrying capacity of blood.
- **Exchange transfusion** : Acute chest syndrome & stroke.
 - Sequential removal of HbS from circulation.
 - Goal : HbS concentration <30% of Hb.
- Hematopoietic stem cell transplant : **Curative.**
- Newer agents : ↓ vaso-occlusive episodes.
 - Crizanlizumab : monoclonal Ab against P-selectin → ↓ attachment of sickled RBCs to vessel wall → ↓ episodes.
 - Voxelotor : ↓ HbS polymerization → ↓ sickling.

Preventive care :

Penicillin prophylaxis : Infancy → 5 yrs of age.

Immunization : Against encapsulated organisms.

Hydroxyurea : ↑ production of HbF → ↓ sickling episodes.

- Started > 9 months of age.

β -THALASSEMIA :

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mode of inheritance :

- Autosomal recessive.

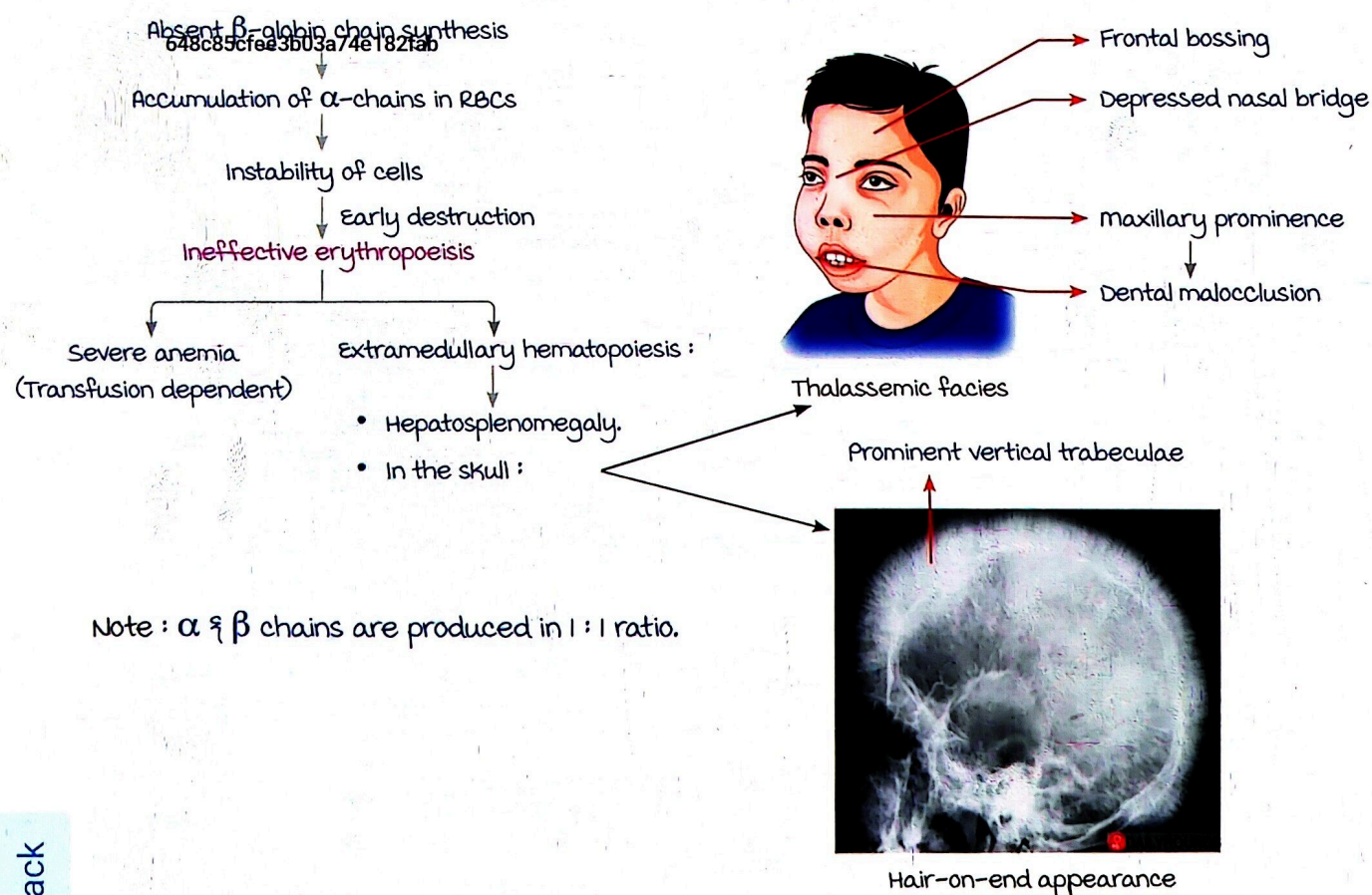
Types :

- Reduction/absence of β -globin chain synthesis.
- Normally, 2 alleles code for β -globin chain.

key :

- β : Normal.
- β^+ : $\downarrow$ production.
- β^0 : No production.

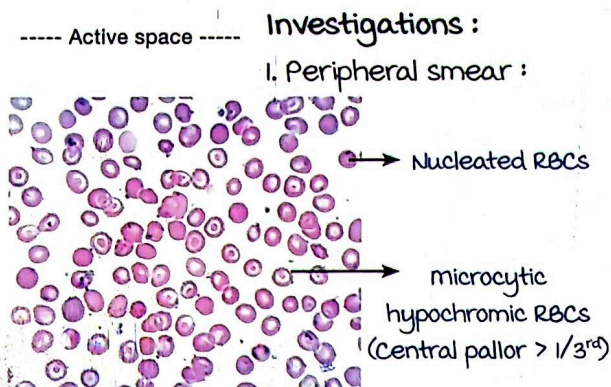
Genotype	Name	Severity
β/β	Normal	-
β/β^+	β -Thalassaemia trait (mild)	Asymptomatic
• β^+/β^+ • β^+/β^0	β -Thalassaemia intermedia	mild to moderate ($\beta^+/\beta^0 > \beta^+/\beta^+$)
β^0/β^0	β -Thalassaemia major/cooley's anemia (most severe)	Severe (Transfusion every 3-4 weeks)

Pathophysiology & clinical features :

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Haematology

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2. Hemoglobin electrophoresis :

- Initial diagnostic test.

3. High-Performance Liquid Chromatography (HPLC) :

- **Gold standard** test.

management :

Supportive :

- Blood transfusion every 3-4 weeks.
 - Leukocyte depleted/irradiated blood products : ↓ risk of allergic reactions.
 - Pre-transfusion target Hb 9-10 g/dL : ↓ es deformities & promote growth.

Curative : Hematopoietic stem cell transplant.

Complication of iron overload :

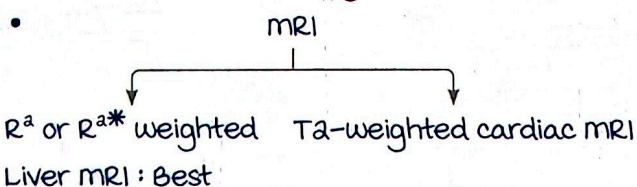
- D/t repeated transfusions (Over 1 yr of transfusion) → monitor.
- major cause of morbidity & mortality (Cardiotoxicity).
- Organs damaged : Liver, spleen, **heart**.
- High transferrin saturation

↓ ses bacteriostatic properties of lactoferrin

↑ risk of infections (m/c : **Yersinia enterocolitica**)

Investigations :

- Serum ferritin > 1000 pg/L.



management :

Iron chelators :

- **Deferasirox** (Oral) : Preferred.
 - Only causes mild GI disturbances.
- **Deferoxamine** (s.c) :
 - Side effects : Ototoxicity & retinal toxicity.
- **Deferiprone** (Oral) :
 - Side effects : Neutropenia & cartilage damage.